

BOAMP Linkage Quality Evaluation

Current two-layer proxy-renewal evidence

Current independent study

End-to-end refresh: 2026-07-28

Every figure below was re-derived from the current artifacts; counts and link identities are unchanged from the 2026-07-27 run, which was verified by the frozen parity fingerprints in `tests/fixtures/baseline_fingerprints.json`.

Abstract

This report audits the current BOAMP proxy-renewal linkage handoff. The reference event is still a constructed linkage event, not verified legal renewal truth. The 2026-07-27 run contains 3,380 eligible sources. Layer 1 produces 10,862 candidate pairs and 1,003 balanced links; Layer 2 produces 13,524 candidate pairs and 1,188 balanced links. The shared balanced threshold is 0.343167. Blocking remains the largest quality constraint, and the duration variable remains a high-risk design input because it affects candidate generation, temporal scoring, and survival modeling.

1 Reference Rule

For each eligible source contract, the pipeline searches later notices from the same buyer identity within the configured temporal window around the estimated contract end date. Candidate scores combine text, CPV, temporal proximity, and buyer-identity reliability:

$$S_{ij} = 0.35s_{\text{text}} + 0.30s_{\text{cpv}} + 0.25s_{\text{time}} + 0.10s_{\text{buyer}}.$$

Only the rank-1 candidate can be accepted. The current Layer-1 percentile thresholds are 0.278387, 0.343167, and 0.442070 for broad, balanced, and strict variants.

2 Blocking and Acceptance

Table 1: Candidate generation and accepted links.

Quantity	Layer 1	Layer 2
Eligible sources	3,380	3,380
Sources with candidates	2,005	2,165
Blocking coverage	59.3%	64.1%
Candidate pairs	10,862	13,524
Balanced links	1,003	1,188
Acceptance rate conditional on candidates	50.0%	54.9%
Overall linking rate	29.7%	35.1%
POTENTIAL-tier links	327	430

Table 2: Layer transition diagnostics.

Transition	Sources
Unlinked in both layers	2,188
Same selected candidate	962
Added by enrichment	189
Changed candidate	37
Removed by enrichment	4

Table 3: Evidence classes. These diagnostics do not create real BOAMP ground truth.

Diagnostic	Layer 1	Layer 2	Interpretation
Fellegi–Sunter precision	0.602	0.815	model-based precision over blocked pairs
Fellegi–Sunter recall	0.212	0.098	model-based recall conditional on blocked pairs
Median link text similarity	0.207	0.191	selected-link lexical support
Median margin	0.113	0.092	ambiguity of best versus second-best candidate

3 Quality Diagnostics

The enrichment-added links are lower-evidence on text: median s_{text} is 0.087 for links added by enrichment versus 0.209 for links shared with Layer 1. That pattern is consistent with the purpose of enrichment—recovering identity-blocking failures—but it also means added links should be interpreted as sensitivity evidence rather than as verified renewals.

4 Survival Handoff

Table 4: Balanced survival comparison.

Quantity	Layer 1	Layer 2
Rows	3,380	3,380
Events	1,003	1,188
Censoring rate	70.3%	64.9%
Survival at 12 months	0.7209	0.6786
Survival at 24 months	0.6908	0.6332
Median survival	Not reached	Not reached
RMST at 60 months	43.51	40.57

5 Main Findings

1. Layer 2 materially increases candidate coverage and accepted proxy events, but it also shifts the evidence mix toward lower text similarity and lower margins.
2. The balanced event remains a proxy. No manual-validation file in the current repository contains completed labels sufficient for a verified precision claim.

3. Blocking, not scoring alone, is the binding ceiling: 1,375 Layer-1 sources and 1,215 Layer-2 sources have no candidate.
4. Duration leakage remains central. Duration determines estimated end dates, temporal blocking, temporal scoring, and then appears as a survival covariate. Any causal reading of the duration hazard ratio is unsupported.

6 Current Source of Truth

The current machine-readable evidence is in:

- data/processed/comparison/layer_comparison_summary.csv;
- data/processed/comparison/layer_link_comparison.csv;
- reports/tables/linkage_fs_precision_recall_by_layer.csv;
- reports/tables/enrichment_added_links_quality.csv;
- reports/tables/survival_layer_conclusions_comparison.csv.